

4. (25 marks) For $(a, b) \in \mathbb{R} \times \mathbb{R}_+$, consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + bx_2^2 + x_3^2) + 2\sqrt{b}x_1(x_2 + x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

Find all the principal submatrices of A .

- (b) (6 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative semidefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative semidefinite?
- (c) (15 marks) Find the values of the parameters a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, plot these regions in $\mathbb{R} \times \mathbb{R}_+$.

5. (30 marks) Let $f : U \rightarrow \mathbb{R}$ be a function defined on a convex subset of $U \subset \mathbb{R}$.

- (a) (10 marks) State a definition of *concavity* for an arbitrary function f . Then, state a definition of *concavity for a twice differentiable function* f , which is based on the Hessian of f .
- (b) (20 marks) Prove that the two definitions you gave in part (b) are equivalent provided that f is twice differentiable.

6. (45 marks) For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & y - e^x \\ \text{s.t.} \quad & ay \leq a^2x + 1 \\ & y \geq 0, x \geq 0. \end{aligned}$$

- (a) (5 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?
- (b) (10 marks) Suppose the Slater condition is satisfied, write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?
- (c) (20 marks) Solve the constrained optimization problem in terms of a .
- (d) (10 marks) State the Envelope Theorem and use it to find the slope of value function with respect to the parameter a . Is the value function concave or convex? For what value of $a > 0$ is the value function minimized?